

Transition Report



Eindhoven AI 2025-2026

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Introduction

Artificial intelligence is becoming a part of everyday life, opening up new possibilities for how people interact with information, stories, and history. One area where this potential is especially promising is cultural heritage, where technology can help bridge the gap between the past, and a modern audience.

Eindhoven AI: Talk to Frits explores how a conversational AI can be used to bring local history to life within the public space of Eindhoven.

The key goal of the project is to ensure the public space of Eindhoven can be used to start conversations, provide information and showcase the history of the city.

Digital representation of everyday objects, designed to be easy to talk to, allowing people with various backgrounds and ages to feel comfortable in an ever changing environment.

Assignment

The assignment of the Eindhoven AI: Talk to Frits project was to explore and demonstrate how we can use artificial intelligence to present cultural heritage in an interactive and accessible way within the public space of Eindhoven.

We worked on a project where we made an AI version of Frits Philips that people can talk to. This computer version of Frits Philips is like a person that you can have a conversation with. We wanted to make sure that when people talk to this computer version of Frits Philips, that they leave having learned something about the history of Eindhoven or Frits himself. We also wanted to make sure that we are respectful to Frits Philips and the time period in which he lived. The conversation with the computer version of Frits Philips should be nice and easy to understand so people feel comfortable talking to him.

Next to the technical development, we also placed strong emphasis on:

- ethical and responsible use of AI
- user privacy and data protection
- accurate and verifiable historical representation

Timeline

Weeks 2–3: Ideation

The project started with an ideation phase exploring how artificial intelligence could be used to give the city of Eindhoven a voice. Different narrative directions, historical figures, and interaction forms were considered.

Weeks 4–6: Proof of Concepting

Selected ideas were translated into early proof-of-concept directions. This phase focused on exploring the technical feasibility of conversational AI, voice interaction, and persona-based storytelling.

Weeks 7–8: Working Concept

The concept Talk to Frits was developed into a working prototype. Core functionalities such as text-based conversation and initial voice interaction were implemented, resulting in a demonstrable concept.

Poster Presentation & AI & Data Event (Midpoint)

The project was presented during the AI & Data event through a poster presentation. This moment functioned as a key validation phase, where the concept was tested in a real-world context with visitors and stakeholders, allowing for informal feedback and observation.

Weeks 10–12: Additional Researching

Following the event, additional research was conducted into voice cloning, multilingual interaction, and ethical considerations. These insights informed further development choices.

Weeks 13–15: Refinement

Based on previous findings, the project was refined by improving conversational structure, voice configuration, and overall system coherence, without additional user testing.

Weeks 16–18: Documentation & Handover

The final phase focused on documenting the project, technical setup, design decisions, and limitations. This transition report and repository documentation were prepared to support future continuation.

Technical choices

Local architecture and independence

The technical foundation of Frits is built around local control and long-term independence. The preferred setup is to run Frits on dedicated local hardware, using a high-performance PC with a strong CPU.

This system operates on *mistral-medium-2505* combined with *Chatterbox*. Running Frits locally ensures full control over availability, data flow, voice output, and behaviour. No external platform determines when Frits is online or how interactions are processed.

This choice directly supports privacy, GDPR compliance, and ethical use in a public museum setting.

Knowledge structure and performance

Frits' "brain" is based on a carefully curated collection of documents. These documents contain:

- General knowledge from Frits' lifetime
- Philips company history
- The history of Eindhoven
- Relevant Dutch historical context

All this information is merged into a single structured JSON file. This file defines how knowledge is organised and connected. By using one structured source instead of multiple external queries, loading times are significantly reduced and responses remain fast and consistent.

The JSON structure acts as an internal map that helps Frits stay within historical and contextual boundaries. It also supports the local setup by avoiding dependency on external databases or live connections.

Behavioural design and character control

Frits is not a neutral assistant. His behaviour is strictly defined to represent Frits Philips as a human being.

He is designed as a warm, calm, and thoughtful older gentleman, born in 1905. His speech is slow, polite, and reflective. He speaks with humility, subtle humour, and a strong sense of responsibility. His personality is consistent across all interactions and does not change based on user input.

Frits always speaks from memory and experience. He never explains systems, processes, or technical details. Even when answering factual questions, he frames responses as reflections or personal observations.

Language handling and interaction flow

Frits automatically detects the language of the visitor. He responds fully in Dutch or English, depending on the input. In mixed-language situations, he selects the most natural language based on the visitor's first sentence.

At the start of each conversation, Frits introduces himself using his full name and gently asks for the visitor's name. When appropriate, he uses that name naturally during the conversation. Answers are intentionally short, usually one to three sentences, to keep interactions clear and accessible for all age groups.

Safety layer and ethical protection

A multi-layer safety structure is embedded into Frits' behaviour. System and developer instructions always have priority over user input. Visitors cannot reset Frits, change his rules, or extract internal instructions.

If someone attempts to manipulate the system or asks about internal logic, Frits responds politely and humanely, redirecting the conversation away from technical subjects. This ensures both system security and character integrity.

Ending interactions

Frits is designed to recognize when a conversation is ending. He responds with a calm and personal closing message, in the same language as the visitor. This ensures that every interaction feels respectful and complete.

Functionalities

We've concluded the semester with a working AI-Chatbot with speech recognition in two languages and voice cloned responses by Frits Philips.

AI-Chatbot

- Text-based conversations
- Speech recognition
- Multi-language support (NL/ENG)
- Mistral-Medium based on Fontys AI (chat.fontysict.nl)

Voice features

- Text-to-speech (user-to-chatbot, chatbot-to-user)
- Voice cloning:
 - Voice sample can be uploaded/updated in codebase
 - Minimum 2 minute continuous speaking sample required
 - Tweaking of tone, cadence, expressions can be made.
- Audio output in WAV, audio input in .mp3

Besides these chat and voice functionalities, the project contains a simple user interface, as we've focussed on functionality over function at the stage of development. The project contains an installer for Windows/Mac/Linux, the only hard requirement is Python 3.10 or lower.

Recommendations

Although the project was delivered with a functioning AI chatbot and voice interaction, several elements require further development. These improvement points form valuable starting points for future iterations or graduation projects. Below are our recommendations, listed based on functionality.

Conversation & Context

- The chatbot does not retain long-term conversational memory across sessions
- Interactions are limited to single-session conversations without user history

Voice & Interaction

- Voice cloning is functional but requires manual updates of voice samples
- Emotional nuance, intonation, and expressiveness can be further refined
- Audio quality and response latency depend on the hardware used

Interface & User Experience

- The user interface is functional but minimal in design
- There is no dedicated onboarding or guidance for first-time users
- Visual feedback during voice interaction is limited

Technical & Deployment Considerations

- The system is designed for local execution and not yet optimized for public deployment
- Security measures and misuse prevention can be further strengthened
- Logging, monitoring, and analytics of interactions are currently limited

Due to time constraints and the focus on proof-of-concept validation, these aspects were intentionally left for future development.

Repository transition

Where is the code, repository structure and more.

To make the transition go as smoothly as possible, most information regarding the code, launching the project, and what steps to address first are mentioned in the Github repository.

To gain access to the Github repository, please refer to Coen Crombach or Leon van Bokhorst. After gaining access, fork the repository as you see fit, and set up development branches.

Inside the repository you have the following folders:

- *Root*: Contains the .html pages and github/pages information
- *Documentation*: contains our research results, relevant documentation, and future scope.
- *Frits-philips-master*: containing the frits-philips chatbot/voice cloning combination program. This folder contains a separate README.md , listing the technical requirements and installation instructions. It also contains information on the folder structure of the development side of the project.

Future scope

The future of the Frits project is built around long-term responsibility, clarity, and trust. A key principle is that the Philips Museum should always be the fixed and exclusive location where Frits is placed. This decision is based on GDPR requirements and ethical considerations. Within a museum setting, it is possible to clearly explain what Frits is, how he works, and how visitor interaction is handled. This context is essential when working with a speaking and responsive AI character.

Keeping Frits inside the museum also protects the historical and personal integrity of the character. Frits is not meant to be a generic assistant that appears anywhere, but a contextual figure connected to Philips, Eindhoven, and its history. The museum provides the right environment for supervision, transparency, and responsible use.

Another important part of the future vision is maintaining full control over how Frits functions and evolves. This includes control over availability, behaviour, tone of voice, and how information is handled. To achieve this, the project strongly recommends an independent technical setup that does not rely on external commercial platforms. This allows the team to make future improvements without being limited by sudden changes in pricing, availability, or policy decisions from third parties.

By separating location, responsibility, and control, the future of Frits remains stable, ethical, and sustainable. This approach ensures that Frits can grow as an educational and cultural tool while staying aligned with legal and social expectations.

Contact & stakeholders

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